ECON2103 Microeconomics

Chapter 6 Production

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Outline

1. Firms and Their Production Decisions
2. Production with One Variable Input
3. Production with Two Variable Inputs
4. Returns to Scale
5. Summary and Exercises

The Production Decisions of a Firm

The production decisions of firms can be understood in three steps:

1. **Production Technology**: How *inputs* (such as labor, capital, and raw materials) can be transformed into *outputs*.
2. **Cost Constraints**:Just as a consumer is constrained by a limited budget, the firm is concerned about the cost of production and the *prices* of labor, capital, and other inputs.
3. **Input Choices**: Given its production technology and the prices of labor, capital, and other inputs, the firm must choose *how much of each input* to use in producing is output.

theory of the firm - Explanation of how a firm makes cost-minimizing production decisions and how its cost varies with its output.

Why Do Firms Exist

- Firms offer a means of *coordination* that is extremely important and would be sorely missing if workers operated independently.
- Firms eliminate the need for every worker to negotiate every task that he or she will perform, and bargain over the fees that will be paid for those tasks.
  - Firms can avoid this kind of bargaining by having managers that *direct the production of salaried workers*—they tell workers what to do and when to do it, and the workers (as well as the managers themselves) are simply paid a weekly or monthly salary.

The Technology of Production

- Firms take *inputs* and turn them into *outputs* (or products).
- **factors of production** - Inputs into the production process (e.g., labor, capital, and materials).
  - We can divide inputs into the broad categories of *labor, materials and capital*, each of which might include more narrow subdivisions.

The Production Function

production function - Function showing the highest output that a firm can produce for every specified combination of inputs.

$$q = F(K, L)$$

- Production functions describe what is *technically feasible* when the firm operates *efficiently*—that is, when the firm uses each combination of inputs as effectively as possible.

The Short Run versus the Long Run

- **short run** - Period of time in which quantities of one or more production factors cannot be changed.
- **fixed input** - Production factor that cannot be varied.
- **long run** - Amount of time needed to make all production inputs variable.

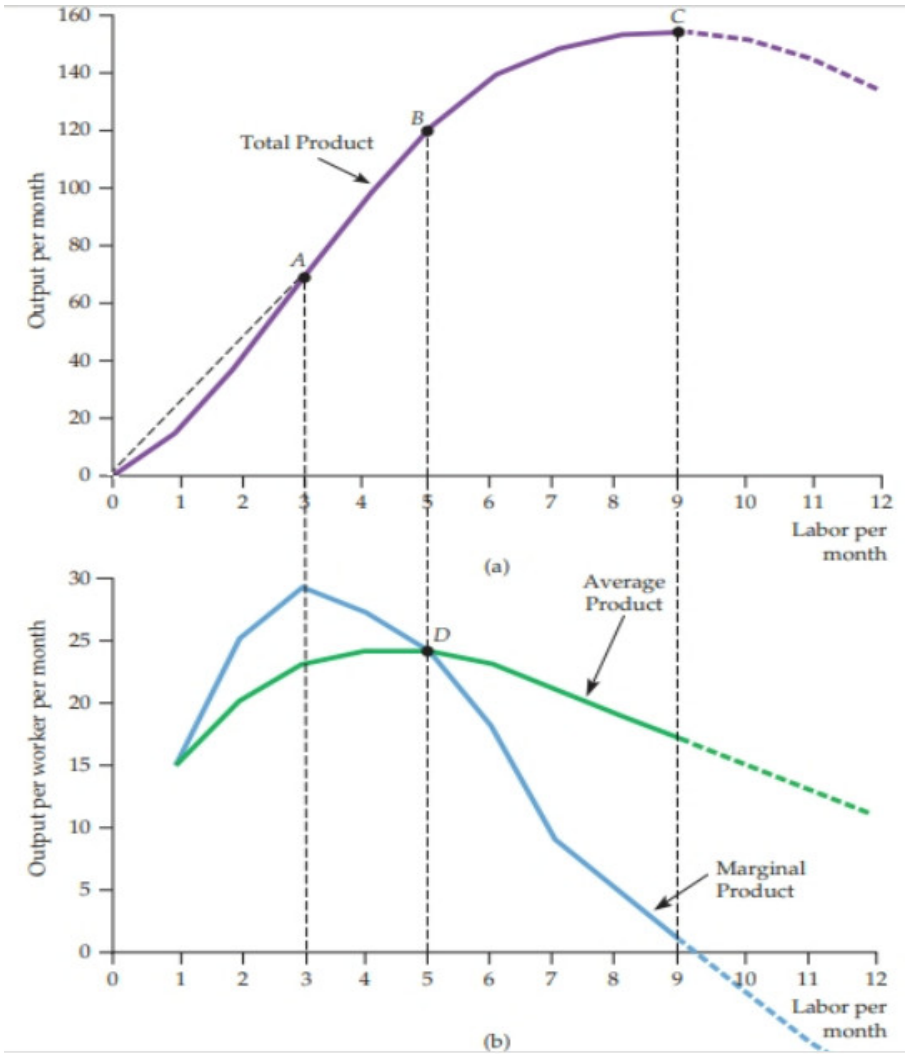
Average and Marginal Products

- **average product** - Output per unit of a particular input
  - Average product of labor = Output/labor input = q/L
- **marginal product** - Additional output produced as an input is increased by one unit
  - Marginal product of labor = Change in output/change in labor input = $\Delta q/\Delta L$

Example: Production with Variable Labor

TABLE 6.1	PRODUCTION WITH ONE VARIABLE INPUT			
AMOUNT OF LABOR (L)	AMOUNT OF CAPITAL (K)	TOTAL OUTPUT (q)	AVERAGE PRODUCT (q/L)	MARGINAL PRODUCT ($\Delta q/\Delta L$)
0	10	0	—	—
1	10	15	15	15
2	10	40	20	25
3	10	69	23	29
4	10	96	24	27
5	10	120	24	24
6	10	138	23	18
7	10	147	21	9
8	10	152	19	5
9	10	153	17	1
10	10	150	15	−3
11	10	143	13	−7
12	10	133	11.08	−10

Slopes of the Product Curve



Slopes of the Product Curve (Con't)

当边际产量（MP）= 平均产量（AP）时，
平均产量达到最大值

- When the marginal product curve is above the average product, the average product is increasing.(Point D)
- When total output is maximized, the slope of the tangent to the total product curve is 0, as the marginal product. (Point C)
- Beyond that point, the marginal product becomes negative and the total output falls as more labor is added.

Relationship Between the Average and Marginal Products

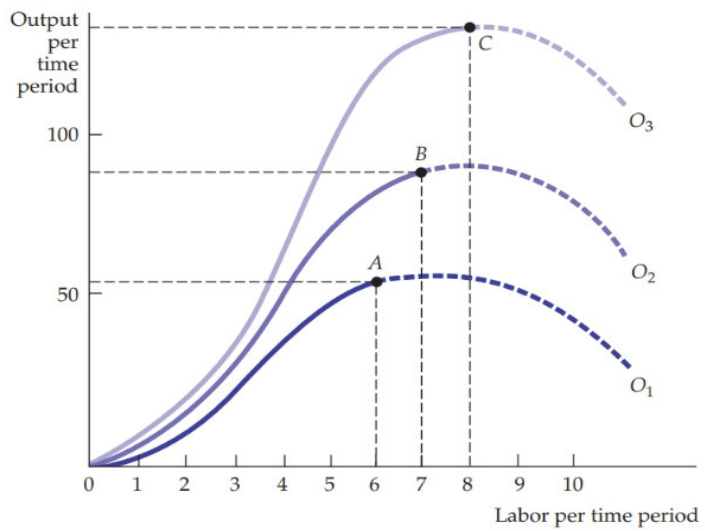
- **The Average Product of Labor Curve**
  - In general, the average product of labor is given by the slope of the line drawn from the origin to the corresponding point on the total product curve.
- **The Marginal Product of Labor Curve**
  - In general, the marginal product of labor at a point is given by the slope of the total product at that point.
- The slope of the tangent to the total product curve at any point between B and C is lower than the slope of the line from the origin.

The Law of Diminishing Marginal Returns

law of diminishing marginal returns - Principle that as the use of an input increases with other inputs fixed, the resulting additions to output will eventually decrease.

The Effect of Technological Improvement

Labor productivity (output per unit of labor) can increase if there are improvements in technology, even though any given production process exhibits diminishing returns to labor.



Labor Productivity

- **labor productivity** - Average product of labor for an entire industry or for the economy as a whole.
- One of the most important sources of growth in labor productivity is growth in the stock of capital
  - **stock of capital** - Total amount of capital available for use in production.
- **technological change** - Development of new technologies allowing factors of production to be used more effectively.

Isoquants

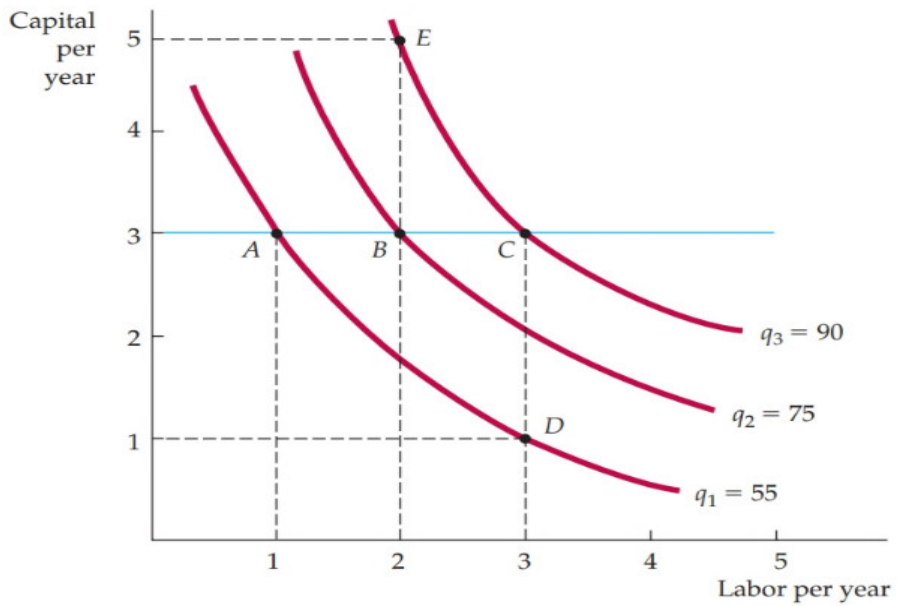
- **Isoquant** - Curve Curve showing all possible combinations of inputs that yield the same output.
- **isoquant map** - Graph combining a number of isoquants, used to describe a production function.

TABLE 6.4 PRODUCTION WITH TWO VARIABLE INPUTS

	LABOR INPUT				
CAPITAL INPUT	1	2	3	4	5
1	20	40	55	65	(75)
2	40	60	(75)	85	90
3	55	(75)	90	100	105
4	65	85	100	110	115
5	(75)	90	105	115	120

Example: Isoquant Map

A set of isoquants, or isoquant map, describes the firm's production function.



Diminishing Marginal Returns

- Isoquants show the flexibility that firms have when making production decisions: They can usually obtain a particular output by substituting one input for another.
- To see why there are diminishing marginal returns to labor, consider the horizontal line ABC.
- Because adding one factor while holding the other factor constant eventually leads to lower and lower incremental output, the isoquant must become steeper as more capital is added in place of labor and flatter when labor is added in place of capital.

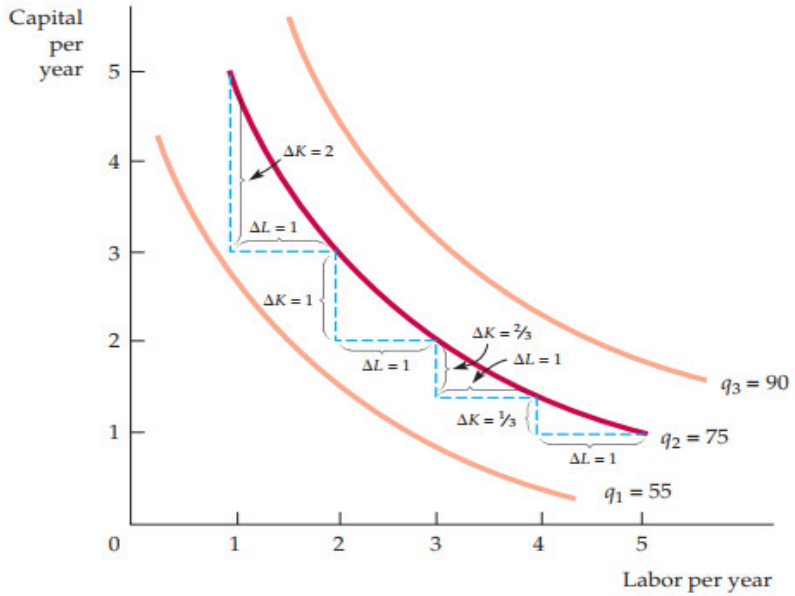
Substitution Among Inputs

MRTS 是等产量曲线（isoquant）在某一点的斜率，
反映了生产要素之间的替代关系。

marginal rate of technical substitution (MRTS) - Amount by which the quantity of one input can be reduced when one extra unit of another input is used, so that output remains constant.

$$\begin{aligned} \text{MRTS} &= \text{Change in capital input/change in labor input} \\ &= -\Delta K/\Delta L(\text{for a fixed level of } q) \end{aligned}$$

Diminishing MRTS



- Like indifference curves, isoquants are downward sloping and convex.
- The slope of the isoquant at any point measures the MRTS of labor for capital.

Diminishing MRTS (Con't)

Additional output from increased use of labor $= (MR_L)(\Delta L)$

Reduction in output from decreased use of capital $= (MP_K)(\Delta K)$

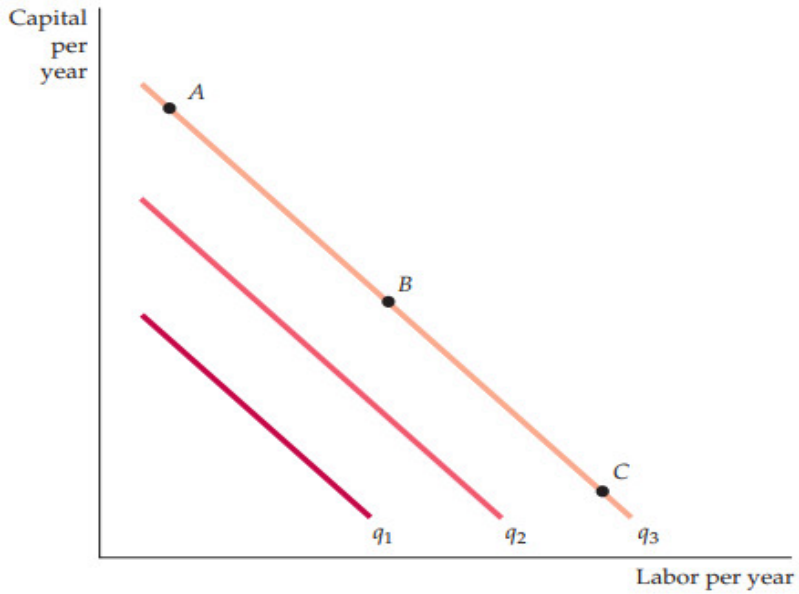
$$(MP_L)(\Delta L) + (MP_K)(\Delta K) = 0$$

By rearranging terms we see that

$$(MP_L)/(MP_K) = -(\Delta K/\Delta L) = MRTS$$

Production Functions - Two Special Cases

Perfect Substitutes

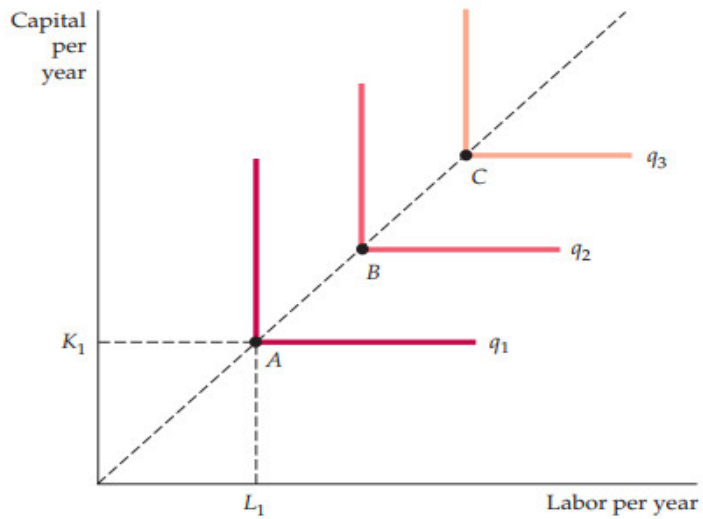


When the isoquants are straight lines, the MRTS is constant. Thus the rate at which capital and labor can be substituted for each other is the same no matter what level of inputs is being used.

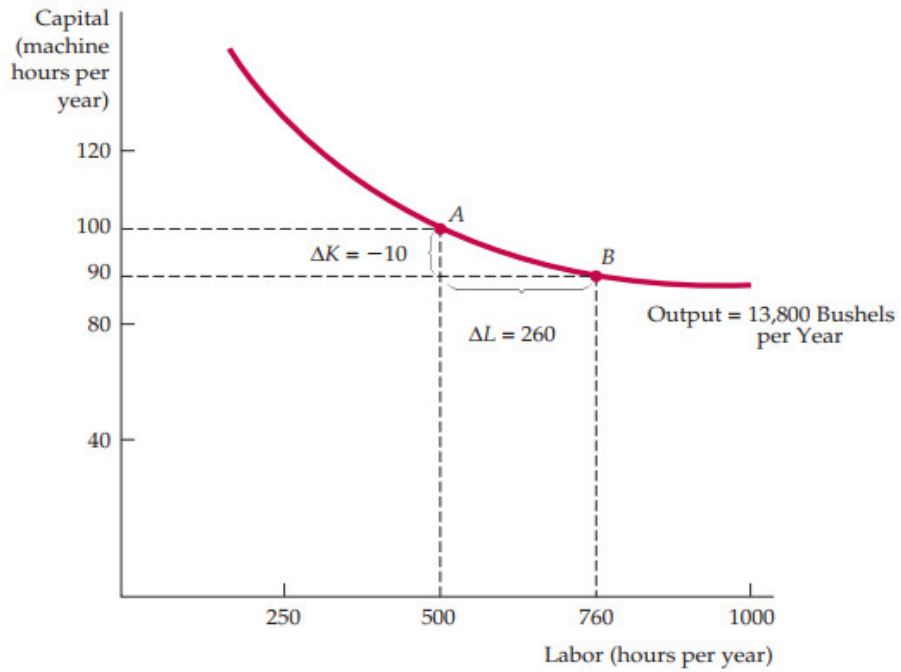
Production Functions - Two Special Cases (Con't)

Fixed-proportions production function

- Production function with L-shaped isoquants, so that only one combination of labor and capital can be used to produce each level of output. It is sometimes called a *Leonitief production function*.
- Describes situations in which methods of production are limited.



Example: A Production Function for Wheat



资本密集型的

The more capital-intensive production process is shown as point A, the more labor-intensive process as point B.

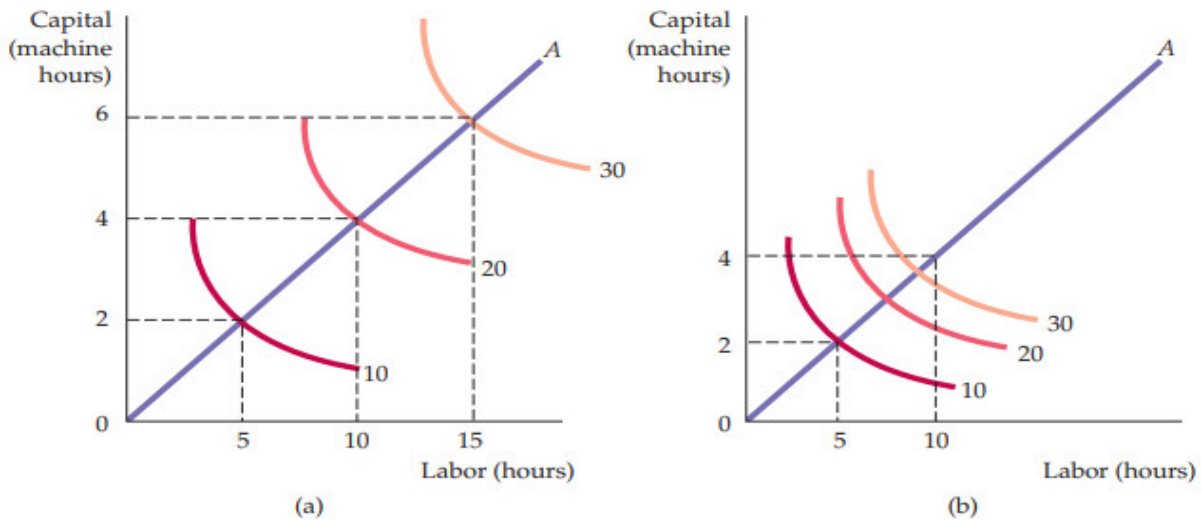
Returns to Scale

returns to scale - Rate at which output increases as inputs are increased proportionately.

- **increasing returns to scale** - Situation in which output more than doubles when all inputs are doubled.
- **constant returns to scale** - Situation in which output doubles when all inputs are doubled.
- **decreasing returns to scale** - Situation in which output less than doubles when all inputs are doubled.

Describing Returns to Scale

- Returns to scale need not be uniform across all possible levels of output. For example, at lower levels of output, the firm could have increasing returns to scale, but constant and eventually decreasing returns at higher levels of output.
- Returns to scale vary considerably across firms and industries. Other things being equal, the greater the returns to scale, the larger the firms in an industry are likely to be.



Summary

- Production function
- Production with one variable input
  - average product and marginal product
  - relationships among total product, average product and marginal product
  - diminishing marginal returns
- Production with two variable inputs
  - isoquant and isoquant map
  - diminishing marginal returns
  - marginal rate of technical substitution
- Returns to scale
  - increasing, constant, and decreasing returns to scale

Exercises

1. We manufacturer automobiles given the production function $q = 5KL$ where q is the number of autos assembled per eight-hour shift, K is the number of robots used on the assembly line (capital) and L is the number of workers hired per hour (labor). If we use $K = 10$ robots and $L = 10$ workers in order to produce $q = 450$ autos per shift, then we know that production is:
   a. technologically efficient.
   b. technologically inefficient.
   c. maximized.
   d. optimal.

Exercises (Con't)

2. When labor usage is at 12 units, output is 36 units. From this we may infer that:
   a. the marginal product of labor is 3.
   b. the total product of labor is 1/3.
   c. the average product of labor is 3.
   d. none of the above.

Exercises (Con't)

3. Assume that average product for six workers is fifteen. If the marginal product of the seventh worker is eighteen,
   a. marginal product is rising.
   b. marginal product is falling.
   c. average product is rising.
   d. average product is falling.

如果 边际产量（MP）大于平均产量（AP）：
平均产量会增加，因为新增单位的产量拉高了平均水平。
如果 边际产量（MP）小于平均产量（AP）：
平均产量会下降，因为新增单位的产量低于平均水平。

Exercises (Con't)

4. Use the following two statements to answer this question:
   - I Isoquants cannot cross one another.
   - II An isoquant that is twice the distance from the origin represents twice the level of output.

   a. Both I and II are true.
   b. I is true, and II is false.
   c. I is false, and II is true.
   d. Both I and II are false.

离原点更远的等产量曲线确实表示更高的产出，但具体倍数关系无法通过距离直接判断。

Exercises (Con't)

5. A production function in which the inputs are perfectly substitutable would have isoquants that are:
   a. convex to the origin. 凸到原点
   b. L-shaped.
   c. linear.
   d. concave to the origin. 凹到原点

Exercises (Con't)

6. According to the diagram below, where each isoquant's output level is marked to the right of the isoquant, production is characterized by:
   a. decreasing returns to scale.
   b. constant returns to scale.
   c. increasing returns to scale.
   d. increasing, constant and decreasing returns to scale.

